MATRIX OPERATIONS · ORDER

AB and BA are not the same object

4
ENTRIES

A 2×3 matrix and a 3×2 matrix can be multiplied in either order, but AB is 2×2 and BA is 3×3. The two products cannot even be compared entry by entry.

A · B

2×3 · 3×2 → 2×2

$$\begin{bmatrix} 5 & 19 \\ 8 & 29 \end{bmatrix}$$

B · A

3×2 · 2×3 → 3×3

$$\begin{bmatrix} 7 & -1 & 19 \\ -4 & 3 & -6 \\ 12 & -6 & 24 \end{bmatrix}$$

ORDER MATTERS Different shapes, so the two products differ before a single entry is compared.

BOTH DEFINED WHEN

m×n and n×m

otherwise only one order conforms

SAME SIZE WHEN

m = n

here 2 ≠ 3, so the entries cannot be compared

SHARED INVARIANT

tr = 34, 34

equal even when the products are not

WHY COMMUTATIVITY FAILS

For square matrices both orders have the same size and the failure shows only in the entries, as in the example above. With different shapes it shows before a single entry is computed.